1. Clinical Question & Core Hypothesis

Clinical Question

Why do some tumors with evidence of epithelial–mesenchymal transition (transitional states, E-cadherin loss) progress to overt pathological invasiveness while others do not, and could microbial triggers (Klebsiella) and TP53 mutation gate this progression via stromal remodeling? 2) Tier 1 paths used: the core mechanistic spine links EMT → invasive cancer and highlights TP53 mutation and E-cadherin (CDH1) as central molecular events that modulate invasiveness; I built the hypothesis on these because they provide a clear, testable molecular mechanism for invasion. 3) Tier 2/supporting evidence: imaging–invasion links (ground glass nodules predict pathological invasiveness), prognostic associations of E-cadherin, and the concept of transitional epithelial–mesenchymal states reinforce a model where cell-intrinsic (TP53, CDH1) and microbe-driven EMT alter tumor–stroma relationships. 4) Endpoint choice: Overall survival (OS) was chosen because it is the most robust, clinically meaningful endpoint available in TCGA-BRCA and will capture the net effect of molecular + microenvironmental drivers of invasion. 5) Exclusions/deliberate simplifications: I excluded explicit inclusion of imaging immune features (TIL/TLS) and dietary/bladder-specific signals (sugary beverages, bladder cancer) from the core hypothesis to keep the mechanistic question focused on EMT–mutation–stroma interactions in tumor invasion; microbial exposure (Klebsiella), CIS, ground-glass imaging correlates and EMT as a process are relegated to secondary_modifiers for follow-up testing?

Working Hypothesis

Does Klebsiella-driven EMT interact with loss of CDH1 and TP53 mutation to promote desmoplastic remodeling and increase stromal-to-tumor ratio (TSR), thereby enhancing invasion in breast cancer? If this interaction exists, tumors with high TSR and concurrent CDH1 loss and TP53 mutation in TCGA-BRCA will have worse overall survival, supporting a microbe–EMT–mutation axis driving aggressive, stroma-rich invasion.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TP53, CDH1
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TP53: TP53 mutation gate this progression via stromal remodeling? 2) Tier 1 paths used: the core mechanistic spine links EMT → invasive cancer and highlights TP53 mutation and E-cadherin (CDH1) as central molecular events that modulate invasiveness; I built the hypothesis on these because they provide a clear, testable molecular mechanism for invasion.

CDH1: CDH1) as central molecular events that modulate invasiveness; I built the hypothesis on these because they provide a clear, testable molecular mechanism for invasion.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	Klebsiella	Supporting modifier from knowledge graph
Secondary	EMT	Supporting modifier from knowledge graph
Secondary	ground glass nodules	Supporting modifier from knowledge graph
Secondary	CIS	Supporting modifier from knowledge graph
Secondary	transitional stages	Supporting modifier from knowledge graph
Secondary	sugary beverages	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: and TP53 mutation synergistically enhance genomic instability and metastatic potential. Microbial triggers can induce EMT and remodel the tumor stroma via inflammatory signaling (e.g., NF-κB, TGF-β), potentially promoting desmoplasia and increasing TSR.

Prediction 2: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Klebsiella: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- EMT: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.